

Weight Considerations in Tactical Missile Design

Conceptual design methods and technologies for tactical missile weight that are addressed in this chapter emphasize low cost and high temperature/short duration structure. Weight considerations include benefits of designing a lightweight missile, subsystem weight sensitivity to flight performance, missile system and subsystem weight prediction, methodology for center-of-gravity and moment-of-inertia prediction, guidelines for factors of safety, applications of micro-machined electro-mechanical systems (MEMS), airframe manufacturing processes and technology, airframe material alternatives, aerodynamic heating prediction, airframe structure/insulation trades, insulation material alternatives, structure design, seeker dome material trades, thermal stress concerns, and areas of localized aerodynamic heating.

Figure 4.1 shows that the next step in the missile synthesis process following propulsion sizing is weight estimation and optimization. Weight estimation and optimization is an iterative process that includes subsystem packaging, weight and balance bookkeeping, and structure weight prediction/optimization.

4.1 Benefits of Lighter Weight Missile

The benefits of designing a lighter weight missile are numerous. They include lower production cost, lower logistics cost, smaller size, higher firepower for a weight-limited/volume-limited launch platform, and reduced observables. The benefits also include better mission flexibility from the capability of carriage on multiple launch platforms (e.g., bombers, fighter aircraft, helicopters, UCAVs, ships, submarines, ground vehicles) and synergy with expeditionary warfare. The rapid force deployment requirement of expeditionary warfare requires lightweight missiles.

4.2 Subsystem Weight Sensitivity to Flight Performance

As shown in Fig. 4.2, the weight of most missile subsystems is sensitive to flight performance requirements such as speed, range, and maneuverability. Missile subsystem weights that are highly impacted by flight performance are propulsion, structure, flight control hardware, surfaces (e.g., wings, stabilizers, flight control surfaces), and aerothermal insulation. Other subsystem weights that have a second-order sensitivity to flight performance requirements are warhead/fusing and power supply. Finally, subsystem weights that are relatively insensitive to flight performance are the seeker dome, seeker, and guidance and control.

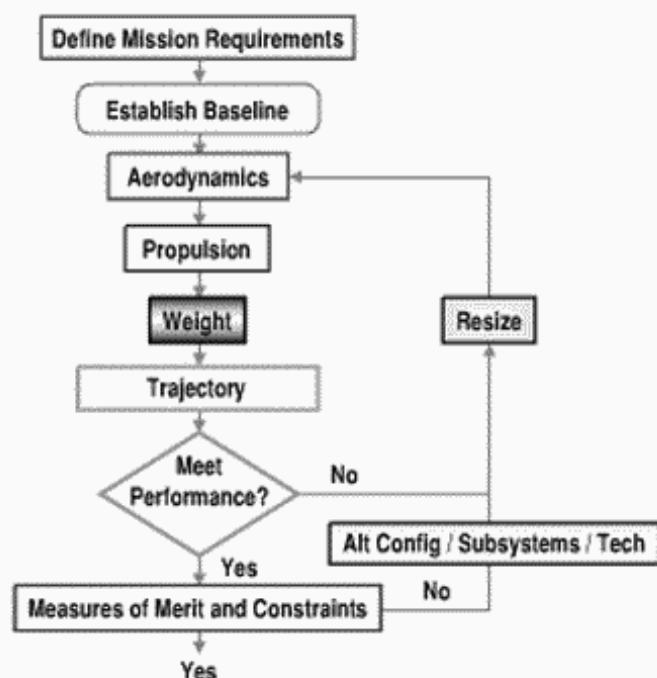


Fig. 4.1 Missile concept synthesis requires iteration.

4.3 Missile Weight Prediction

The average subsystem density of a tactical missile is about 0.05 lb/in.^3 . Except for the nose section, most missile bodies can be represented as a cylinder of diameter d and length l . The volume of a cylinder is $V = (\pi/4)ld^2$. Neglecting the geometry difference of the missile nose from that of a cylinder and neglecting the weight of the surfaces, the missile weight should correlate as the average density times the volume of a cylindrical body [$W_L = \rho_M V = 0.05(\pi/4)ld^2 = 0.04ld^2$].

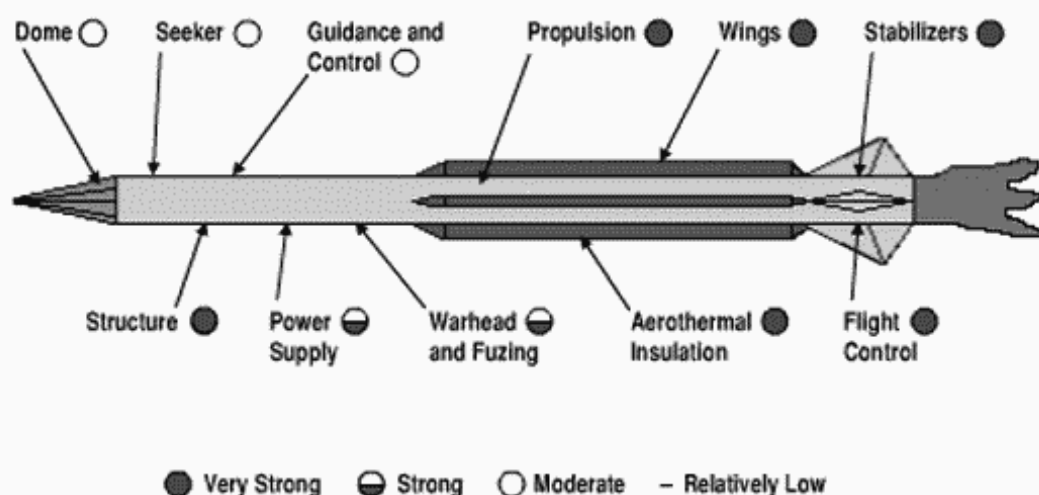


Fig. 4.2 Subsystem weight is sensitive to flight performance (range, speed, maneuverability).

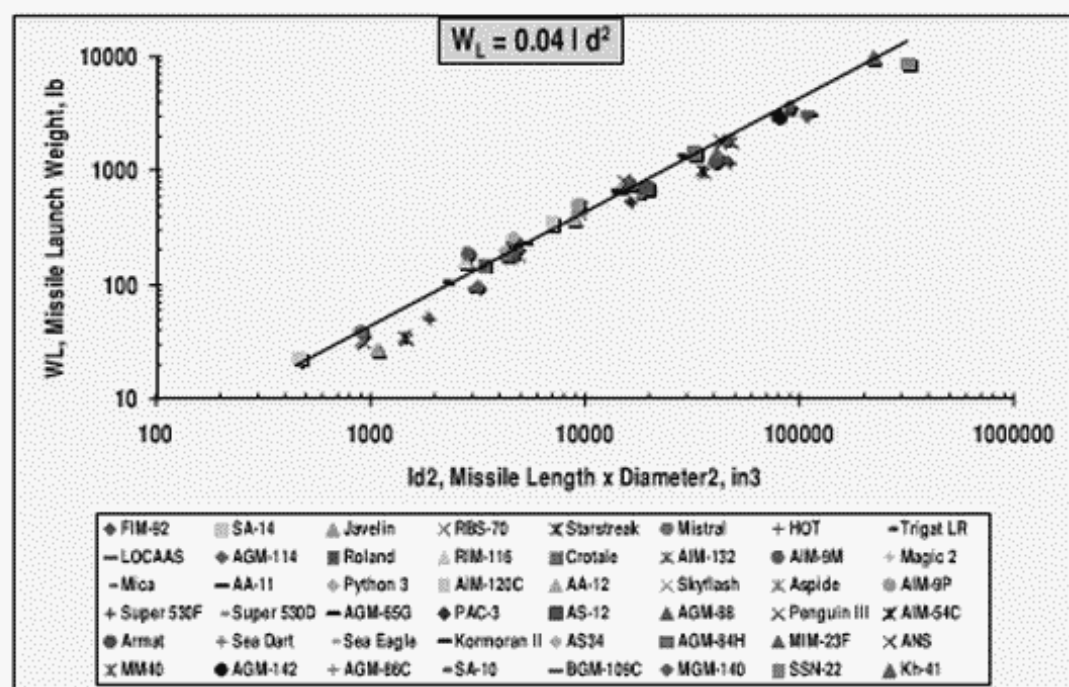


Fig. 4.3 Missile weight is a function of diameter and length.

Figure 4.3, which is based on the weight of 48 tactical missiles, shows the correlation of the predicted weight with the actual weight. The scatter in the data includes the effects of differences in the launch environment, maneuverability, subsystem density, nose fineness, body cross-sectional geometry, boattailing, and the surface planform geometry. Note from the figure that the lightest weight missiles (FIM-92, SA-14, Javelin, RBS-70, Starstreak, Mistral) are man-portable, tube launched missiles with a relatively benign launch environment, relatively small warhead, and small surfaces.

A more detailed description of missile weight at the subsystem level is shown in Fig. 4.4. Shown are typical values of subsystem density for rocket-powered missiles. Except for structure density, the subsystem density varies from 0.04 to 0.07 lb/in.³ A more accurate estimate of missile weight is obtained by estimating the individual subsystem volumes, densities, and weights through a missile weight statement.

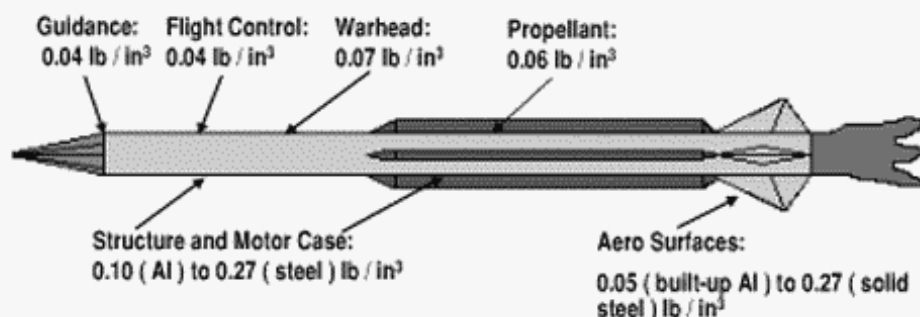


Fig. 4.4 Subsystems densities for rocket-powered missiles.

4.4 Center-of-Gravity and Moment-of-Inertia Prediction

The methodology for missile weight, balance, and moment-of-inertia prediction is shown in Fig. 4.5. Missile weight is the summation of the individual weights of the subsystems, structure, surfaces, and fuel/propellant. Launch weight and burnout weight estimates are required for analysis of the missile flight performance.

The missile center of gravity is estimated by summing each subsystem weight times its distance from the nose tip, divided by the missile weight. As an example, the longitudinal location of the center of gravity is

$$x_{CG} = \Sigma (x_{\text{subsystem}1} W_{\text{subsystem}1} + x_{\text{subsystem}2} W_{\text{subsystem}2} + \dots) / W_{\text{total}}$$

Again, x_{CG} must be calculated for launch and burnout.

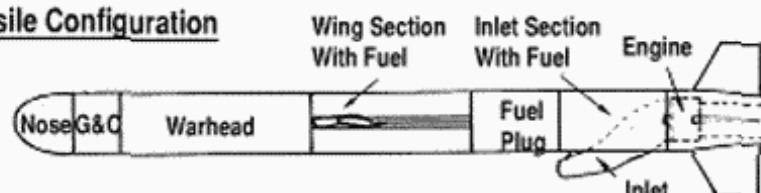
The moment of inertia about the missile center of gravity is calculated by summing the local moment of inertia of each subsystem about its center of gravity, plus the weight of each subsystem times the square of the distance of the subsystem from the missile center of gravity. As an example, the moment of inertia about the y axis is

$$I_y = \Sigma [(I_{y,\text{subsystem}1})_{\text{local}} + W_{\text{subsystem}1} (x_{\text{subsystem}1} - x_{CG})^2 / g_c + (I_{y,\text{subsystem}2})_{\text{local}} + W_{\text{subsystem}2} (x_{\text{subsystem}2} - x_{CG})^2 / g_c + \dots]$$

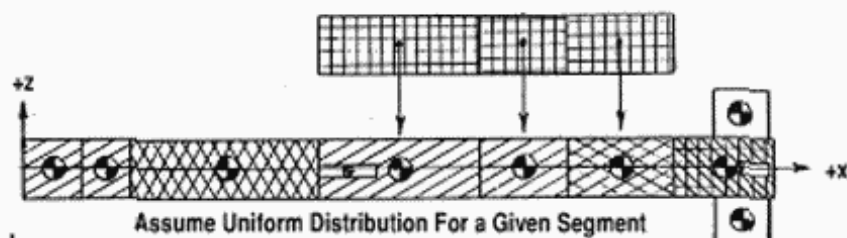
Moment of inertia is calculated for both the launch and burnout conditions.

Figure 4.6 shows the local moment of inertia $(I_y)_{\text{local}}$ variation with body fineness ratio l/d for cylindrical and conical bodies. Note the large increase in the moment of inertia with increasing fineness ratio. As an example, the ramjet baseline can be approximated as a cone-cylinder configuration for the moment-of-inertia calculation. The local moment of inertia of the conical nose about its center of gravity is

Example Missile Configuration



Model



Legend

	Structure and Subsystems		Engine Structure and Subsystems
	Warhead and Structure		Fuel
	Inlet Structure and Subsystems		Aero Surfaces

$$x_{CG} = \Sigma (x_{\text{subsystem}1} W_{\text{subsystem}1} + x_{\text{subsystem}2} W_{\text{subsystem}2} + \dots) / W_{\text{total}}$$

$$I_y = \Sigma [(I_{y,\text{subsystem}1})_{\text{local}} + W_{\text{subsystem}1} (x_{\text{subsystem}1} - x_{CG})^2 / g_c + (I_{y,\text{subsystem}2})_{\text{local}} + W_{\text{subsystem}2} (x_{\text{subsystem}2} - x_{CG})^2 / g_c + \dots]$$

Fig. 4.5 Modeling weight, balance, and moment of inertia.

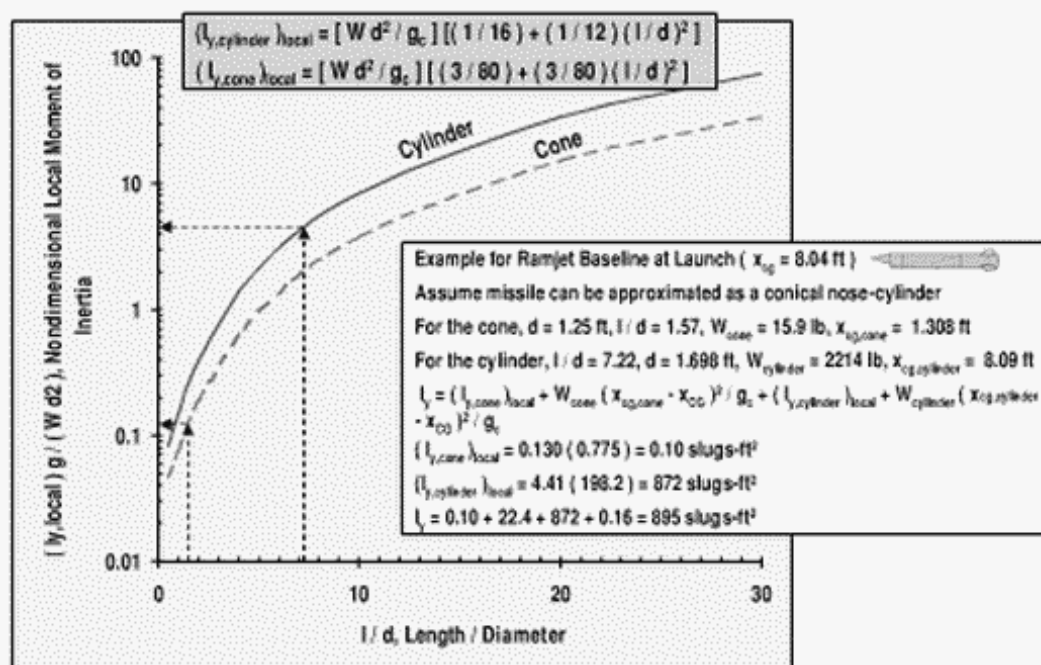


Fig. 4.6 Missile moment of inertia is higher for a high fineness ratio body.

$(I_{y,cone})_{local} = 0.13 \text{ slugs-ft}^2$. The local moment of the cylindrical approximation of the rest of the missile is $(I_{y,cylinder})_{local} = 872 \text{ slugs-ft}^2$. The moment of inertia about the missile center of gravity is given by

$$I_y = (I_{y,cone})_{local} + W_{cone}(x_{cg,cone} - x_{CG})^2/g_c + (I_{y,cylinder})_{local} + W_{cylinder}(x_{cg,cylinder} - x_{CG})^2/g_c$$

Substitution gives the total moment of inertia of the ramjet baseline at launch as $I_y = 895 \text{ slugs-ft}^2$.

4.5 Factor of Safety

Typical factors of safety for tactical missiles are shown in Fig. 4.7. The factor of safety of a missile tends to be high where there is a danger of fratricide. As an example, pressure bottle ultimate and yield factors of safety are typically 2.5 and 1.5, respectively. Missile gas bottles may be pressurized up to 10,000 psi. Because gas bottles require periodic logistics maintenance and inspection by ground personnel, the required factor of safety is high. Another area where the factor of safety is high is in the area of ground handling loads, such as cross-country transportation. Factors of safety for ground handling loads are about 1.5 for ultimate loads and 1.15 for yield loads. Other examples of high factor of safety are launch platform carriage and separation. During carriage or during aircraft separation, missile factors of safety are required to be about 1.5 for ultimate and 1.15 for yield. The motor case is designed not only for conditions of environmental extremes, such as a hot day, but also for consideration of safety. The ultimate and yield factors of safety for the motor maximum effective operating pressure are about 1.5 and 1.1, respectively. The required factors of safety are lower for flight conditions where the missile is safely away from the launch platform. For example, missile-free flight

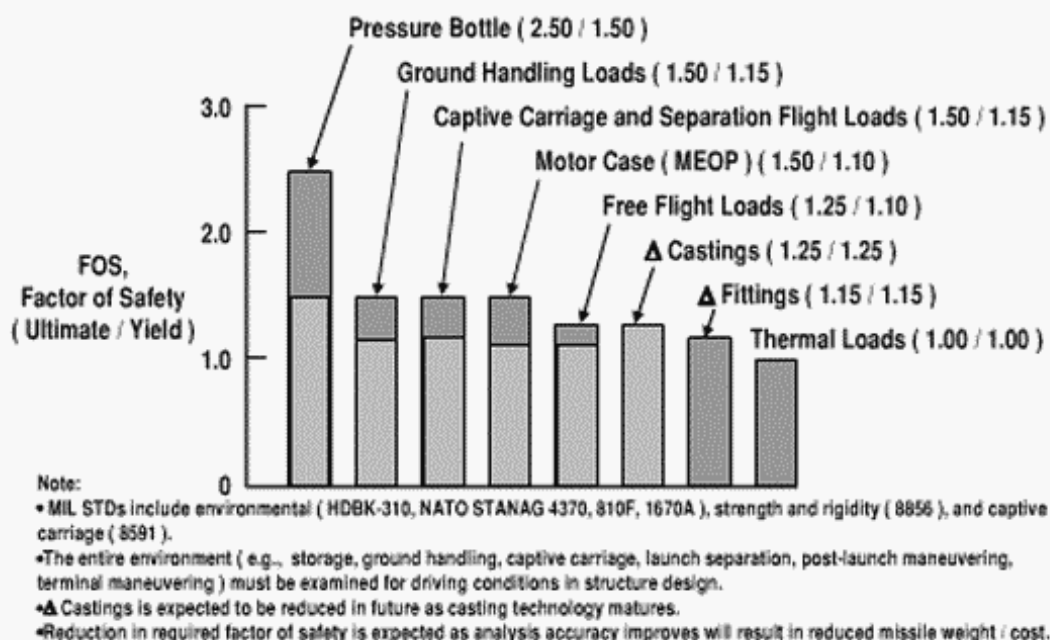


Fig. 4.7 Structure design criteria factor of safety for missiles.

load factors of safety are about 1.25 and 1.1, respectively, and the thermal loads, which occur near the end of flight, are just design considerations with a factor of safety of 1.0. A distinguishing characteristic of tactical missiles is a lower factor of safety compared with manned aircraft or even unmanned air vehicles (UAVs). Because missiles are a throwaway, the factor of safety can be reduced if there is no danger involved to friendly forces. This would result in lighter weight compared with an aircraft or a UAV. Note that an additional factor of safety is required for structural areas where there is a relatively large design uncertainty. An example is castings, which can have hidden voids. Castings currently require an incremental factor of safety of about 1.25 in addition to the normal design factors of safety. Fittings also require an additional factor of safety of about 1.15 because of the uncertainty in the analysis for attachment integrity. The applicable military standards in the United States that are considered in factors of safety include environmental (HDBK-310, NATO STANAG 4370, MIL-STD-810F, MIL-1670A), strength and rigidity (MIL-STD-8856), and captive carriage (MIL-STD-8591) military standards.

A technology opportunity to reduce missile weight is to develop more accurate predictions of the missile environment, loads, and strength. This would allow a reduction in the factor of safety (i.e., factor of ignorance).

4.6 Micro-Machined Electro-Mechanical Systems (MEMS)

Because high performance missiles such as ramjets are weight and volume limited, there is leverage in reducing weight. Reduced weight is consistent with reduced cost, improved performance, and improved launch platform compatibility. One approach to reduce missile weight is to reduce the required factors of safety. Technology in improved analysis and development tools will provide reductions in missile weight and cost by reducing the design uncertainty and the required factors of safety. An example is the MEMS technology illustrated in Fig. 4.8.

◆ **Micro-machined Electro-Mechanical Systems (MEMS)**

- ◆ Small size / low cost
- ◆ Semiconductor manufacturing process
- ◆ 2,000 to 5,000 sensors on a 5 inch silicon wafer

◆ **Missile Development Application**

- ◆ Wireless (RF) Data Collection and Health Monitoring

◆ **Distributed Sensors Over Missile**

- ◆ Stress / strain
- ◆ Vibration
- ◆ Acoustics
- ◆ Temperature
- ◆ Pressure

◆ **Allows Reduced Design Uncertainty / Factor of Safety**

- ◆ Provides reduced weight and cost

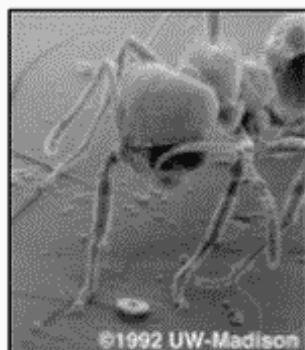


Fig. 4.8 Small-size MEMS sensors can reduce required factor of safety, saving missile weight.

MEMS devices are fabricated from a single piece of silicon by semiconductor manufacturing processes, resulting in a small size, low-cost package. For example, between 2000 and 5000 MEMS sensor devices can be produced from a single 5-in. silicon wafer. Future tactical missiles will have low-cost/small size MEMS sensors for data collection during missile development and for health monitoring after production. Localized stress/strain, vibration, acoustics, temperature, pressure, and other measurements can be monitored through many sensors distributed around the airframe. The higher confidence because of MEMS data will allow weight reduction in the overdesigned structure.

4.7 Manufacturing Processes

Figure 4.9 shows examples of manufacturing processes that are used for tactical missile airframes and surfaces. A driving consideration for cost reduction is to minimize the parts count. Examples of manufacturing processes that reduce the parts count include vacuum assisted resin transfer molding (RTM), filament winding, pultrusion, casting, vacuum bag/autoclave forming, metal forming, strip laminate, and compression molding.

Examples of low-cost manufacturing processes that are particularly applicable to complex shapes are precision casting, vacuum assisted RTM, filament winding, and pultrusion. Precision casting is particularly suitable. It has high payoff for reducing the cost of high temperature metal airframes with complex shape. A historical limitation in applying castings to complex configurations is the tight manufacturing tolerances required for the complex configurations. However, new technology such as ceramic tooling allows low-cost precision castings suitable for complex airframe configurations such as ramjets. Castings reduce the parts count with a resulting cost savings. Large precision cast structures are in development for complex missile shapes, such as ramjets. A one-piece cast airframe design integrates

Geometry Alternatives	Structure Concept Alternatives	Structure Manufacturing Process Alternatives									
		Composites						Metals			
		Vacuum Assist RTM	Compression Mold	Filament Wind	Pultrusion	Thermal Form	Vacuum Bag / Autoclave	Cast	High Speed Machine	Forming	Strip Laminate
Lifting Body Airframe 	Monocoque	●	●	●	●	●	●	●	—	●	
	Integrally Hoop Stiffened	●	●		●	●	●	●	—	●	
	Integrally Longitudinal Stiffened	●	●		●		●	●	—	●	
Axisymmetric Airframe 	Monocoque	●		●	●		●	●	●	●	●
	Integrally Hoop Stiffened	●					●	●	●		
	Integrally Longitudinal Stiffened	●			●		●	●	●		
Fin or Wing 	Monocoque	●	●			●	●	●	●	●	
	Sandwich	●	●				●	●		○	

Note: Manufacturing process cost is a function of recurring cost (unit material, unit labor) and non-recurring cost (tooling).

Note: ● Very Low Parts Count ○ Low Parts Count ○ Moderate Parts Count — High Parts Count

Fig. 4.9 Structure concepts and manufacturing processes for low parts count.

all of the secondary structure to minimize parts count. Precision casting minimizes subsequent machine and hand finishing of mating surfaces by achieving a precision surface finish "as-cast." Fuel cells can be an integral part of the structure and not require bladders. Structural attachment points (e.g., ejector attachments, payload supports, booster attachments) and self-indexing/aligning features can be integral to the structure. This minimizes or eliminates mating/alignment/assembly tooling and test/inspection requirements. Precision castings have been demonstrated for missile aluminum, titanium, and steel airframes, motor cases, and combustors.

Shown in Fig. 4.10 are schematics of the manufacturing processes for vacuum assisted RTM, composite filament winding, composite pultrusion, and metal casting. As stated previously, these processes are particularly suited to lifting body airframes with complex shapes, providing cost benefits from the reduced parts count.

The vacuum assisted resin transfer molding process consists of inserting fibers into a mold, pulling a vacuum, injecting resin into the fibers, and curing the part. The vacuum assisted RTM process has advantages of high production rate and good tolerances (e.g., ± 0.005 in.).

Filament winding is an automated process that is suited to circular and elliptical cross-sectional shapes such as airframes and motor cases. The filament fibers are pulled through a resin bath and wound around a mandrel. The resin is usually worked into the fibers by roll coaters or by breaker bars in dip tanks. Cycle time is about 1 h for winding and about 3 h for oven curing. Because the winding action itself compacts the laminate, vacuum bagging or other compaction methods are not necessary.

Pultrusion is a continuous, automated closed-molding process that is cost effective for high volume production of circular and elliptical cross-sectional shapes. The process pulls the fiber and resin continuously through a heated steel die, where the part cures.

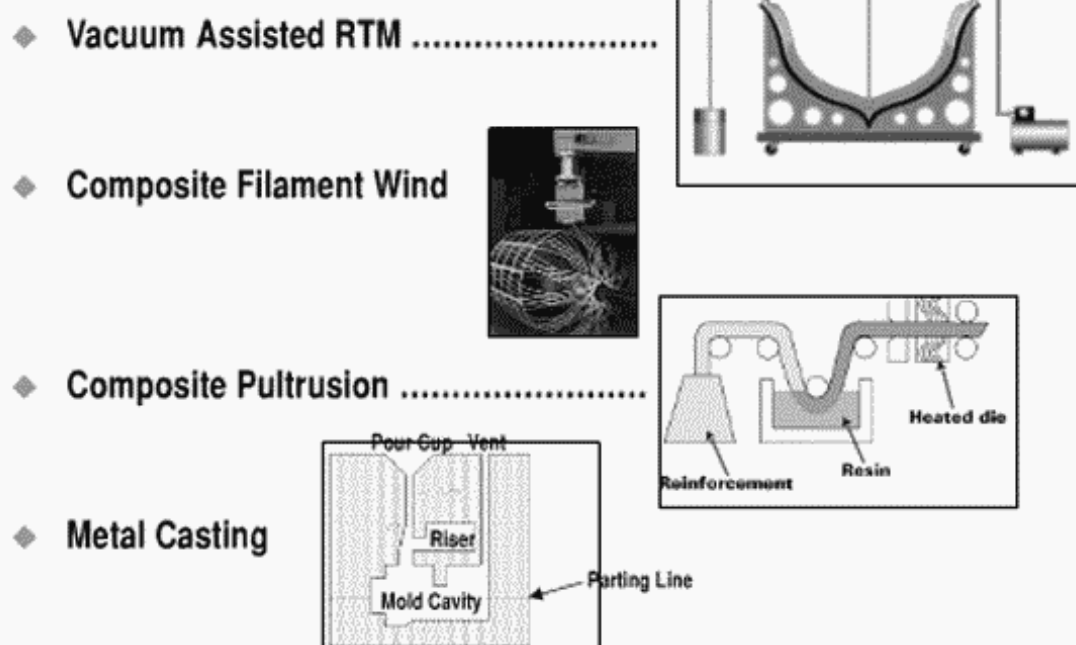


Fig. 4.10 Low parts count manufacturing processes for lifting body airframe.

Metal casting is particularly suited to reducing the parts count and cost of complex, nonaxisymmetric airframes, such as ramjets. It is also applicable to surfaces. Precision casting is based on a permanent precision die, such as a ceramic die. The mold cavity is filled with molten metal by gravity pouring. As the metal cools, the mold cavity, riser, and pour cup become one solid casting. When the metal has cooled and solidified, the ceramic die is opened at the parting line and the part is removed by techniques such as vibration and water blasting.

4.8 Airframe Material Alternatives

Examples of tactical missile alternative airframe materials and their tradeoffs are shown in Fig. 4.11. Metallic airframe material alternatives of Aluminum 2219, Steel PH 15-7Mo, and Titanium 6Al-4V are shown in the order of increasing cost. Similarly, composite airframe material alternatives of 1) S994 glass/epoxy and S994 glass/polyimide, 2) glass or graphite reinforced molding, and 3) graphite/epoxy and graphite/polyimide are shown in the order of increasing cost. Design considerations for airframe materials are maximum allowable stress, compression stability, maximum temperature for the short-duration flight of tactical missiles, sensitivity to thermal stress, ease of joining to adjacent structure/subsystems, low cost, and light weight.

For a lightweight tactical missile airframe at moderate temperatures, graphite/epoxy and graphite/polyimide materials may be preferred. For a high temperature airframe, titanium or steel materials are often preferred. For a low-cost airframe the preferred materials are usually aluminum, steel, S994 glass/epoxy, or S994 glass/polyimide.

Figure 4.12 shows the strength of typical airframe materials. Shown are the tensile strength and the stress-strain curves of small diameter graphite, Kevlar,

Type	Material	Tension (F_{TU}/ρ)	Compression Stability (F_{CR}/ρ)	Max Short-Life Temp	Thermal Stress	Joining	Cost	Weight
Metallic Increasing Cost ↓	Aluminum 2219	○	●	—	—	●	●	○
	Steel PH 15-7Mo	●	—	●	○	●	●	—
	Titanium 6Al-4V	●	○	●	●	○	—	○
Composite Increasing Cost ↓	S994 Glass / Epoxy and S994 Glass / Polyimide	●	○	○	●	○	●	●
	Glass or Graphite Reinforce Molding	—	○	○	●	○	○	●
	Graphite / Epoxy and Graphite Polyimide	●	○	●	●	—	—	●

Note: ● Superior ● Above Average ○ Average — Below Average

Fig. 4.11 Tactical missile airframe material alternatives.

and glass fibers (without a composite matrix); high strength steel (such as precipitation hardening stainless steel); titanium alloy; and aluminum alloy material alternatives. Note that the strength capability of advanced small diameter fibers is very high. For example, as shown in the figure, the unidirectional tensile strength of graphite (carbon) fiber is more than 400,000 psi. Small diameter Kevlar and glass fibers may have comparable strength to graphite or carbon fiber, but also exhibit relatively high strain under load. In addition to small diameter fibers, advanced

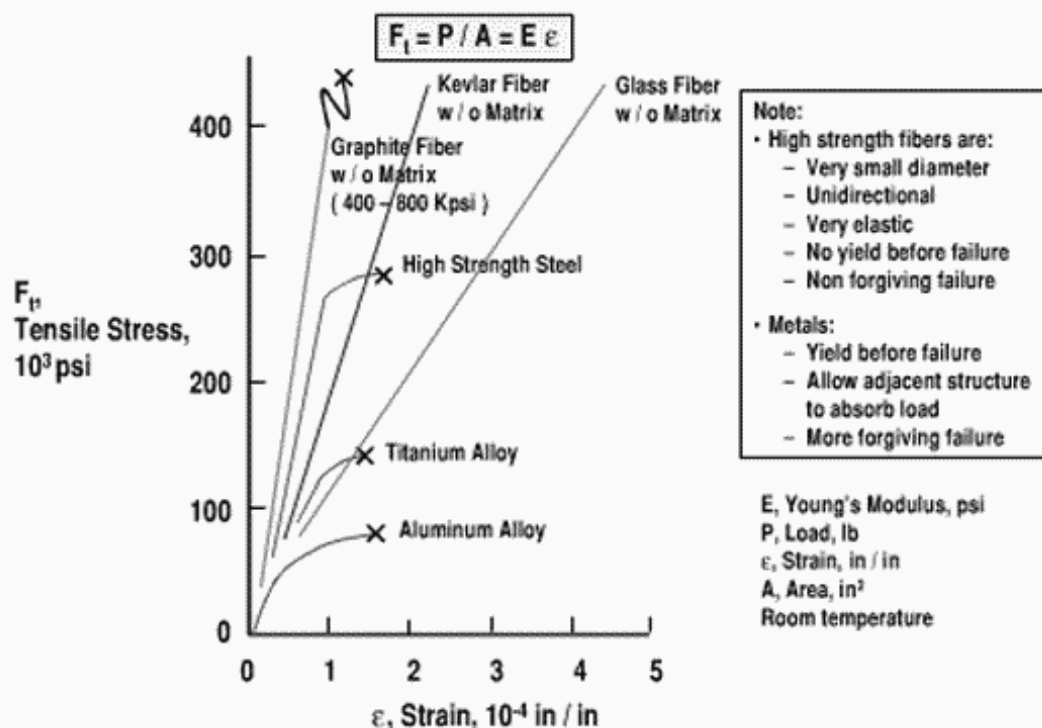


Fig. 4.12 Strength, elasticity of airframe material alternatives.

composite structures have long, continuous fibers and a fiber/matrix ratio that is greater than 50% fibers by volume. Fibers can be graphite (carbon), Kevlar, glass, boron, ceramic, silicon carbide, quartz, polyethylene, and others. As an example of strength at the structure level, a 50% volume graphite composite structure can have strength in a tailored laminate that is above 200,000 psi, much greater than that of aluminum, or even most steels. Also, the low density of composites further reduces the weight compared with metals. Graphite fiber composite materials have extremely high moduli of elasticity, resulting in low strain and deflection compared to metals. However, note with caution that composites are not fault-tolerant. The fiber has a nearly linear stress-strain curve out to the ultimate load, at which it suddenly fails. Unlike metals that generally yield gracefully before ultimate failure, composite fibers generally fail suddenly without yield. Another disadvantage of composite structure is that failure of the structure may be hidden inside the laminate. Microcracks can spread through the matrix, decreasing the ability of the fiber to transfer load.

Figure 4.13 shows the structural efficiency advantage of composites compared to conventional materials. The example is based on a $0\pm45\text{--}90\text{-deg}$ orientation tailoring of the laminate. For short duration temperatures up to about 300°F , graphite/epoxy is a good candidate material based on its characteristics of high strength and low density. Graphite/polyimide can be used at higher temperatures, up to about 1100°F short duration temperature. Epoxy and polyimide matrices are thermoset polymer matrices. Thermosets have relatively high temperature

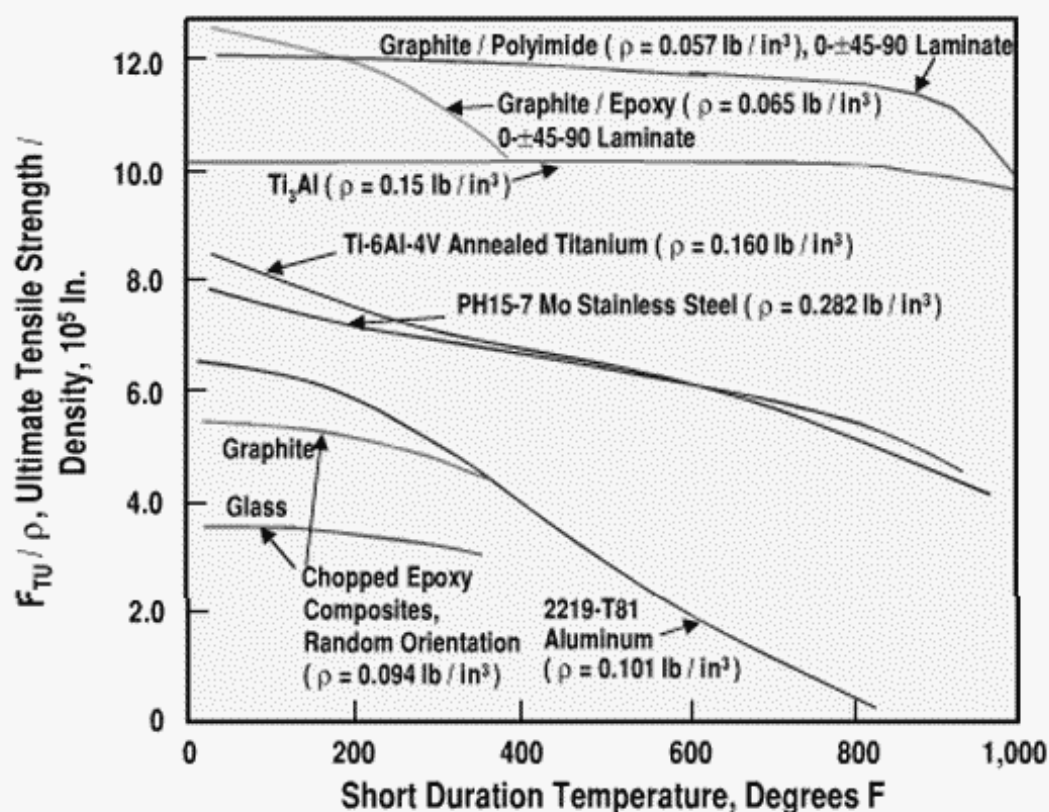


Fig. 4.13 Structural efficiency at high temperature of short-duration airframe material alternatives.

capability, good dimensional stability and solvent resistance, but they require a relatively long cure time. A common high temperature thermoset resin is phenolic. Phenolics are based on phenol (carbolic acid) and formaldehydes. An alternative low-cost polymer matrix with a shorter processing time is thermoplastic. Thermoplastics are most suitable to subsonic missiles because of a relatively low temperature capability. Examples of thermoplastic polymer matrices include polyetheretherketone (PEEK), polyphenylene (PPE), and polyetherketone (PEK). Above 1100°F, titanium and steel alloys are the best materials based on strength-to-weight ratio.

A consideration in developing a design based on composites is complexity. In addition to the material itself having greater complexity, the structures are not as straightforward as metal structures. The designer of a metal structure has two primary design considerations, material choice and the geometric configuration. Composites require another design consideration, the composite matrix orientation. Two structures of identical geometric configuration, weight and composite material, but with different layup, will have different stress-strain and failure modes as a function of orientation. The failure modes of composites, which include exploding laminate, fibers pulling free from a matrix, first-ply failure, matrix cracking, and delamination, are difficult to predict.

Metal matrix composites and ceramic matrix composites are in development for strategic missile, space, UAV, and aircraft applications. An example is titanium matrix/silicon carbide. Although the performance is very high, cost is a factor. Metal matrix and ceramic matrix composites may not be cost effective for tactical missiles.

Examples of uninsulated structure materials that are cost effective for the short-duration flight of tactical missiles are shown in Fig. 4.14. An assumption is that

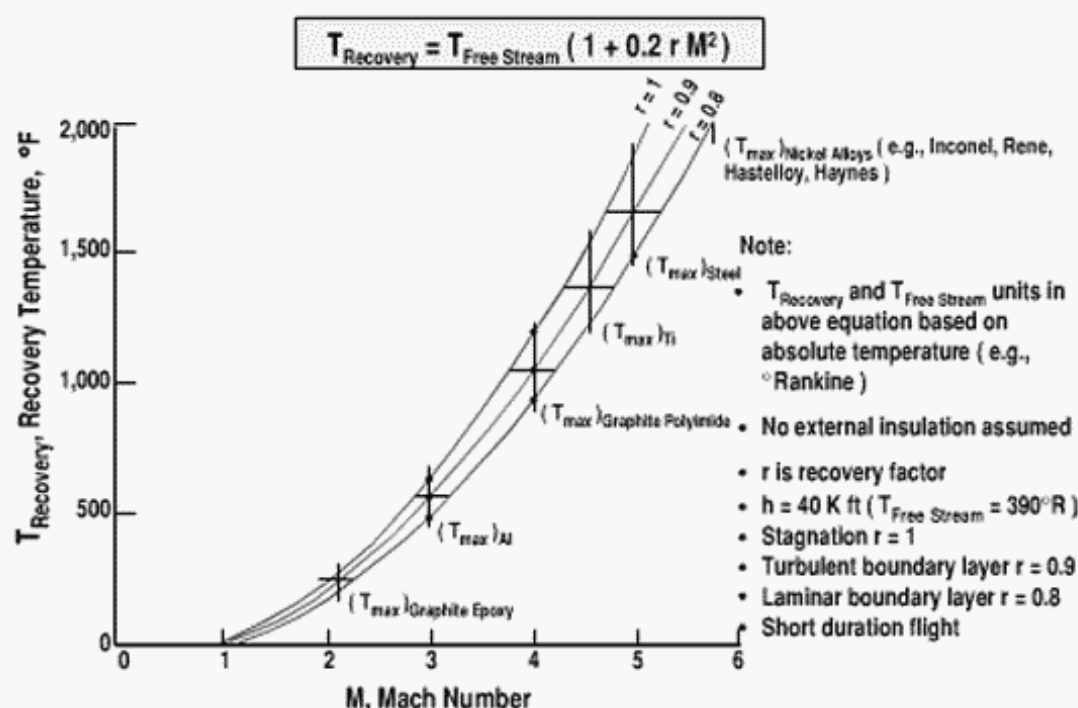


Fig. 4.14 Hypersonic missiles require high temperature structure.

the heat conducted to the airframe is large compared to the heat radiated by the airframe. This assumption is most applicable at low altitude/high atmospheric density. The airframe temperature would be lower at high altitude, due to the relative importance of radiation from the airframe. The example airframe materials selected for the figure are based on the consideration of weight, cost, and maximum temperature capability. Composite materials are a new technology that will find increased use in new missile airframe structure. High temperature composites have particular benefits for hypersonic missiles, providing weight reduction. Titanium alloy technology also enables lighter weight missiles in a hypersonic, high temperature flight environment.

As shown in the figure, at subsonic and low supersonic Mach number, graphite/epoxy and aluminum or aluminum alloys are attractive choices for a lighter weight structure without external insulation. Graphite/epoxy and aluminum alloys have high strength-to-weight ratio, are easily fabricated, have good corrosion resistance, and are low cost. For higher Mach number, graphite/polyimide composite structure has an advantage of high structure efficiency at higher temperature for Mach numbers to about Mach 4. For flight to about Mach 4.5 without external insulation, titanium structure and its alloys are preferred. A disadvantage of a titanium structure is higher material and machining cost. For example, a titanium part has a material cost that is up to 18 times that of aluminum and a machining cost that is up to 13 times that of aluminum. However, the cost to cast a part made of titanium is comparable to the cost to cast an aluminum part. Small tolerance (e.g., ± 0.001 in.) is required to avoid expensive touchup machining. Up to Mach 5.7 without external insulation (about 2000°F), super nickel alloys such as Inconel, Rene, Hastelloy, and Haynes must be used. Precision casting should be used to minimize the expensive machining and material cost associated with super alloys. Above Mach 5.7 the super alloys require either external insulation or active cooling. Active cooling is usually not cost effective for tactical missiles.

The Mach number and temperature application relationships shown in the figure are somewhat dependent upon the temperature recovery factor. At a stagnation region, such as the nose or leading edges, the recovery factor is about 1, resulting in the highest (stagnation) temperature. A turbulent or laminar boundary layer downstream of the nose or leading edge will have temperature recovery factors of about 0.9 and 0.8, respectively, with local temperatures less than stagnation.

4.9 Aerodynamic Heating Prediction

The heat flux input due to aerodynamic heating shown in Fig. 4.15 is a function of Mach number, altitude, and location from the leading edge. It is based on Ref. 11. The equation is $Q = 345\rho^{0.8}M^{2.8}/x^{0.2}$. Note that the heat transfer rate increases with higher Mach number, lower altitude, and shorter distance from the leading edge. The results are based on the assumptions of a turbulent boundary layer, a local angle of attack of 0 deg, and insulation behind the airframe (i.e., adiabatic wall).

An example is shown in the figure of the heat transfer rate of the rocket baseline missile. For an assumed distance of 1 ft from the leading edge, the heat transfer rate of the rocket baseline at Mach 2, 20,000 ft in altitude is 12 Btu/ft²/s.

An example of the skin temperature rate of increase for the rocket baseline airframe is shown in Fig. 4.16. The figure is based on Ref. 11. Note from the

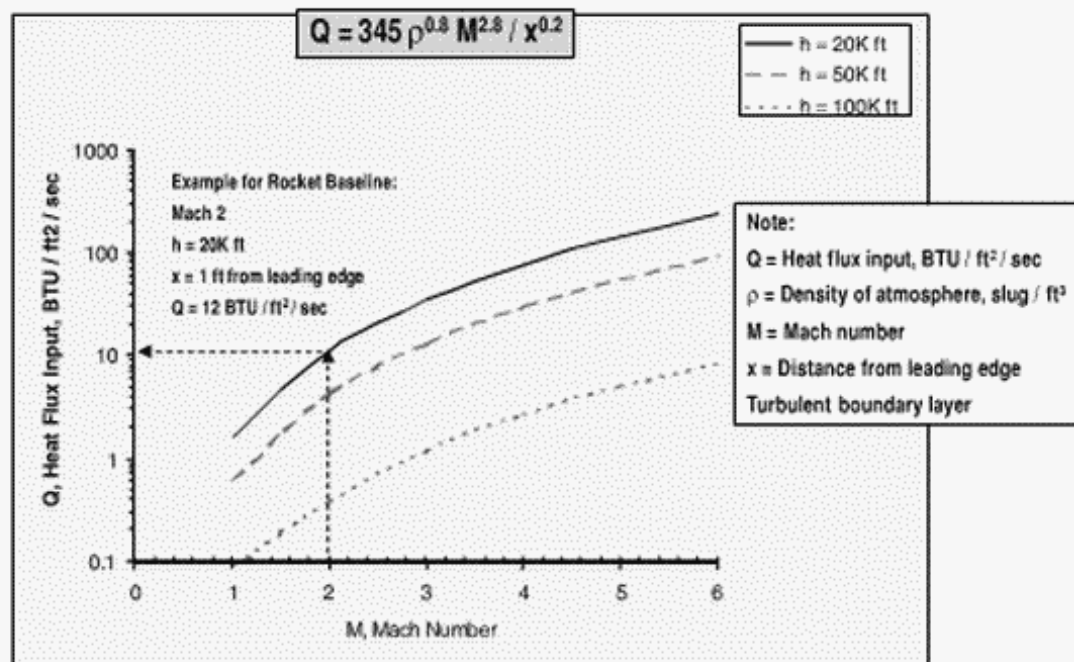


Fig. 4.15 Hypersonic missiles at low altitude have high aerodynamic heating (from Ref. 11).

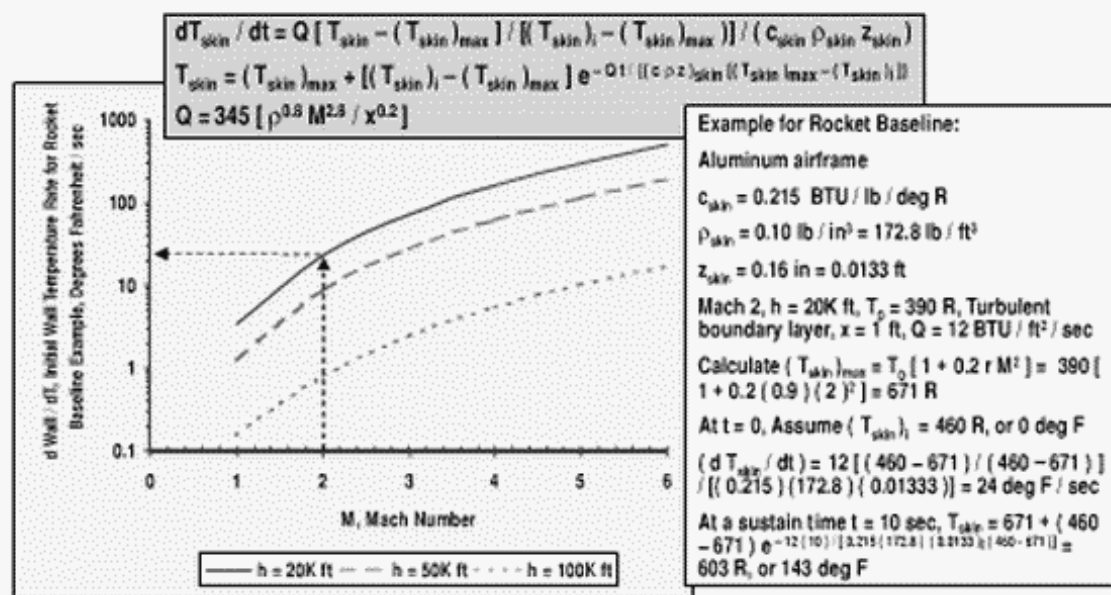


Fig. 4.16 High temperature rate occurs at high Mach number and low altitude (from Ref. 11).

equation

$$dT_{\text{skin}}/dt = Q\{[T_{\text{skin}} - (T_{\text{skin}})_{\text{max}}]/[(T_{\text{skin}})_i - (T_{\text{skin}})_{\text{max}}]\}/(c_{\text{skin}}\rho_{\text{skin}}z_{\text{skin}})$$

that the airframe skin temperature rate is a function of heat flux input, skin temperature, skin conductivity, skin density, and skin thickness. Also note from the figure that the skin temperature rate is highest at high Mach number and low altitude. The assumptions for the rocket baseline missile are an aluminum airframe, airframe skin thickness of 0.16 in., and insulation behind the airframe skin (i.e., adiabatic wall). As shown previously, at Mach 2 and 20,000 ft in altitude, the heat flux input at 1 ft behind the leading edge is 12 Btu/ft²/s. For this flight condition, the initial skin temperature rate is computed to be 24°F/s.

The equation for airframe skin temperature as a function of time is

$$T_{\text{skin}} = (T_{\text{skin}})_{\text{max}} + [(T_{\text{skin}})_i - (T_{\text{skin}})_{\text{max}}]e^{-Qt/[(c\rho z)_{\text{skin}}(T_{\text{skin}})_{\text{max}} - (T_{\text{skin}})_i]}$$

For a short duration time of flight, the rocket baseline airframe will not reach thermal equilibrium. As an example, the maximum temperature at Mach 2, 20,000 ft in altitude for a turbulent boundary layer ($r = 0.9$) is

$$(T_{\text{skin}})_{\text{max}} = T_0[1 + 0.2rM^2] = 390[1 + 0.9(2)^2] = 671^\circ\text{R}$$

Assuming an airframe skin initial temperature of $(T_{\text{skin}})_i = 460^\circ\text{R}$, the skin temperature after 10-s exposure is $T_{\text{skin}} = 603^\circ\text{R}$, or 143°F.

4.10 Insulation Trades

Thermal insulators are used to provide short-duration protection of structural materials from aerodynamic and propulsion heating. Airframe structure/insulation trades for tactical missiles include hot structure/internal insulation, external insulation/"cold" structure, and a one-piece self-insulating composite structure. These are shown in Fig. 4.17. A consideration for a volume-limited missile is the total thickness of the structure/insulation. Large thickness means less volume for fuel or propellant, resulting in less range. An uninsulated metal structure has the smallest thickness at low Mach number/stagnation temperature. For a relatively short duration flight at very high Mach number/stagnation temperature, an external insulator/ablator over a metal airframe with internal insulation is the lightest weight approach, at the expense of large thickness. The external insulation maintains the airframe below its maximum allowable temperature, and the internal insulation maintains subsystems (e.g., warhead, propellant/fuel, electronics) below their maximum allowable temperature. At intermediate (supersonic) Mach number, the lightest weight approach for a relatively short duration flight is either an uninsulated metal airframe with internal insulation (to protect the subsystems) or a self-insulating composite structure.

The figure also shows typical values of the insulation parameters for the alternative insulation approaches. Values of maximum temperature T_{max} , thermal conductivity k , heat capacity c , density ρ , and diffusivity α for a typical metal structure (aluminum), composite structure (graphite/polyimide), internal insulation (Min-K), and external insulation (microquartz paint) are shown. As an external insulator, microquartz paint provides a high temperature capability ($T_{\text{max}} = 2000^\circ\text{F}$). It protects a low temperature airframe such as aluminum ($T_{\text{max}} = 600^\circ\text{F}$)

Example Structure / Insulation Concepts	T_{max}	k	c	ρ	α
Metal Structure Without Insulation (e.g., Aluminum)	600	0.027	0.22	0.101	0.000722
Metal Structure	600	0.027	0.22	0.101	0.000722
Internal Insulation (e.g., Min-K)	2000	0.0000051	0.24	0.012	0.00000106
Composite Structure (e.g., Graphite Polyimide)	1100	0.000109	0.27	0.057	0.00000410
Ext Insulation (e.g., Micro-Quartz Paint)	1200	0.0000131	0.28	0.012	0.00000226
Metal Structure	600	0.027	0.22	0.101	0.000722
Internal Insulation	2000	0.0000051	0.24	0.012	0.00000106

Note:

- Small thickness allows more propellant / fuel for diameter constrained missiles (e.g., VLS launcher)
- Weight and cost are application specific
- T_{max} = max temp, deg F; k = thermal conductivity, BTU / sec / ft / deg F; c = thermal capacity, BTU / lbm / deg F; ρ = density, lbm / in³; α = thermal diffusivity = $k / (\rho c)$, ft² / sec

Fig. 4.17 Structure/insulation trades.

in a short-term, high temperature environment. Similarly the low conductivity of Min-K ($k = 0.0000051$ Btu/s/ft²/°F) as an internal insulator protects warhead, electronics, and other subsystems. As a combined structure and insulator material, graphite/polyimide has good characteristics of maximum temperature ($T_{max} = 1100^{\circ}\text{F}$) and thermal conductivity ($k = 0.000109$ Btu/s/ft²/°F). The thickness that is required for insulation is a design consideration. Thermal diffusivity is a measure of the insulation thickness efficiency of a material. It is a function of the thermal conductivity, heat capacity, and density of the material, given by the equation $\alpha = k/(\rho c)$. As a comparison, the thermal diffusivity of Min-K is 0.1% that of aluminum. Microquartz paint and graphite/polyimide also have low thermal diffusivity compared with aluminum (0.3% and 0.6% respectively).

For environmental control of a tactical missile, an actively cooled (refrigerator) approach is usually not cost effective. Passive insulation using thermal inertia is usually a superior approach for the short-duration flight of a tactical missile. In the case of a ramjet, the fuel can provide a thermal sink.

4.11 Insulation Material Alternatives

Because missiles have volume and weight constraints, higher density insulation materials are in development. Higher density insulation materials permit more fuel/propellant for a volume-limited missile, resulting in longer range. Figure 4.18 shows the maximum temperature and short-duration insulation efficiency of candidate insulation materials.

Note that composite materials are good candidates for lightweight insulation. For insulating high-speed airframes, engines, and motor cases, the medium density plastic composites are often used. An example is a fiberglass-reinforced phenolic resin containing nylon, silica, graphite, or carbon. Phenolics have good resistance to erosion, allow high surface temperatures over 5000°R, and exhibit good

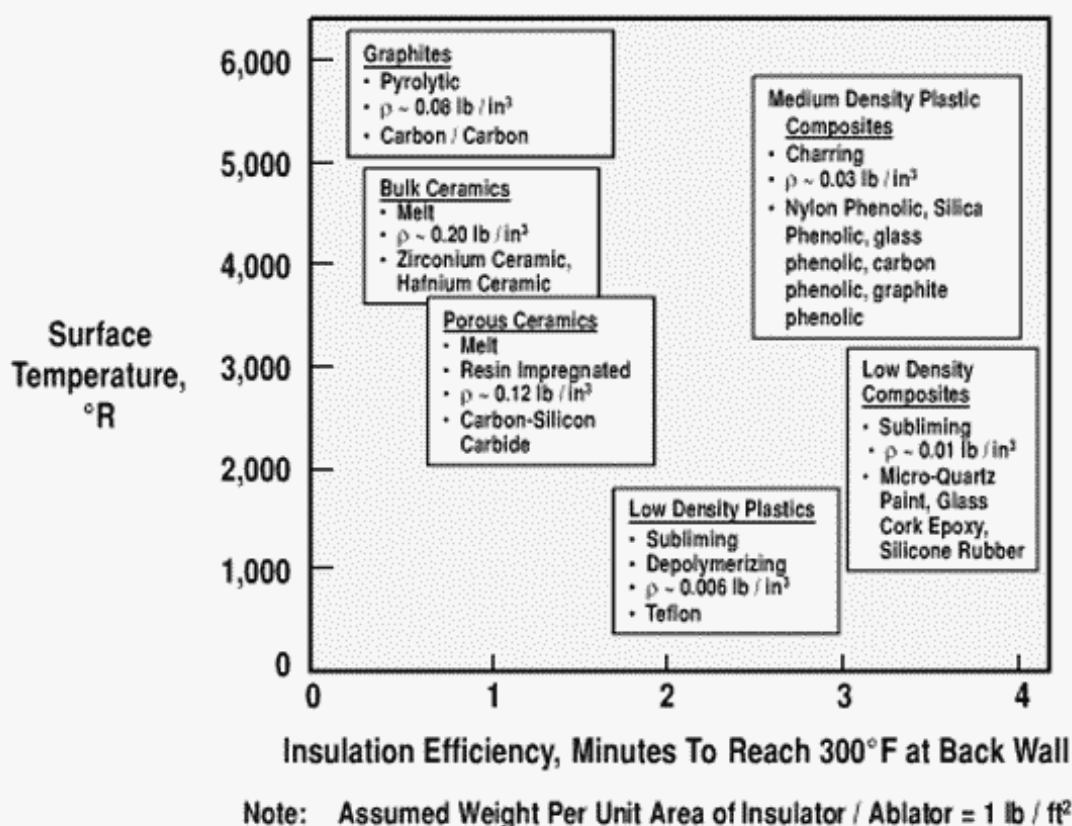


Fig. 4.18 Composites are good insulators/ablaters at high temperature.

insulation performance. These materials char at high temperature, but they generally maintain their thickness and aerodynamic shape. They are usually fabricated by wrapping fiberglass tape over a metal form mandrel, so that the grain of the finished unit is oriented for minimum erosion. Another example of a good insulator at somewhat lower temperatures is low density composites. Low density composites such as microquartz paint, glass cork epoxy, or silicone rubber may be used for temperatures up to about 3000°R. These require periodic maintenance touch-up. A third approach based on lower density plastics is rarely used for hypersonic missiles. Low density plastics have a disadvantage of decomposing at high temperature into gas, resulting in decreased thickness and changes in the aerodynamic shape.

Ceramic refractory materials and graphite materials are also candidate insulators for high speed airframes, engines, and motor cases. Although ceramic refractory materials and graphites have high temperature capability, the insulation efficiency for a given weight of material is not as good as that of plastic composite materials. An example of a porous ceramic, with a maximum temperature up to about 3500°R, is resin impregnated carbon/silicon carbide. At high temperatures the resin melts, providing cooling for the structure. Examples of bulk ceramics are zirconium ceramic and hafnium ceramic. Bulk ceramics are capable of withstanding temperatures up to 5000°R, but like porous ceramics, they have relatively poor insulation efficiency. Finally, graphite insulators provide the highest temperature capability. Graphites are capable of withstanding temperatures greater than 5000°R. However, graphites have relatively poor insulation efficiency.

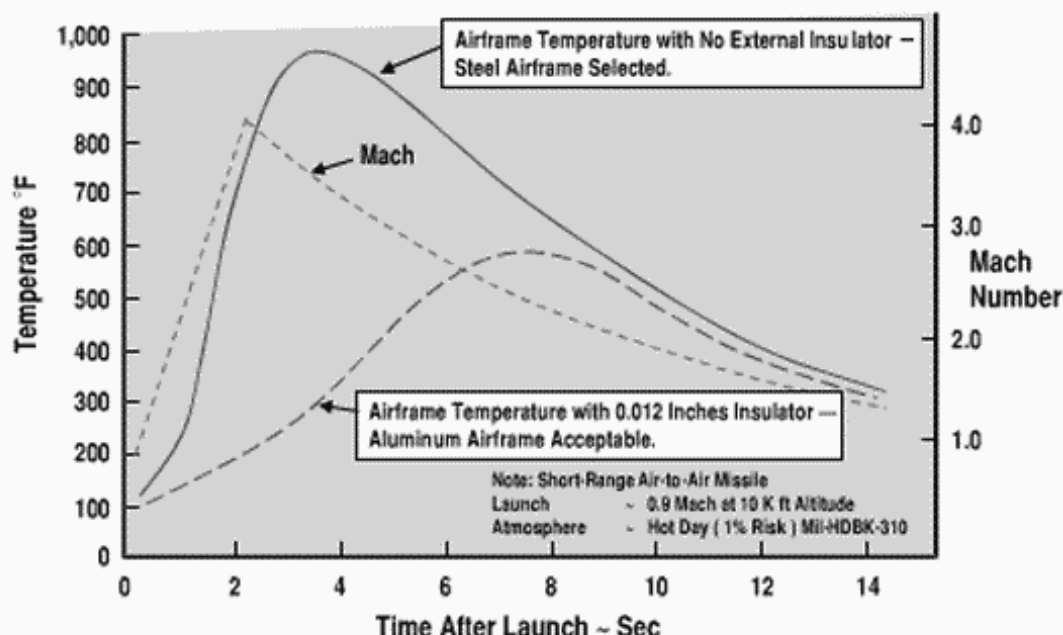


Fig. 4.19 External insulation has high payoff for short-duration flight.

Figure 4.19 compares the temperature for an insulated airframe vs an uninsulated airframe for short-duration flight. The example is based on a short-range air-to-air missile launched on a hot day at Mach 0.9 and 10,000 ft in altitude. The missile motor burnout is at 2.5 s, providing a burnout Mach number of 4.0. Note from the figure that for an uninsulated airframe a steel structure was selected because of the high maximum temperature (950°F). An alternative is to use a lighter weight, lower cost aluminum airframe that is protected by a thin layer of spray-on insulation. The peak temperature on the insulated structure is only 600°F, allowing the use of aluminum in the short-duration flight at high Mach number.

4.12 Structure Design

Localized buckling, bending moment loads, and motor case pressure loads are structure design considerations for tactical missiles. The equation for localized buckling stress due to bending is $\sigma_{\text{buckling, bending}}/E \cong 0.35(t/r)$. The equation for localized buckling stress due to compression is $\sigma_{\text{buckling, axial compression}}/E \cong 0.25(t/r)$. Localized buckling stress in bending and compression is shown in Fig. 4.20 as a function of thickness ratio and modulus of elasticity. Note that for small thickness the allowable stress for localized buckling is very low. Localized buckling may occur in a thin-wall structure prior to exceeding the yield stress.

An example is shown of the required motor case wall thickness to avoid localized buckling. It is based on the bending moment of the rocket baseline missile. The rocket baseline motor case is 4130 steel, with a modulus of elasticity $E = 29.5 \times 10^6$ psi. The case thickness $t = 0.074$ in. and the case radius $r = 4$ in. For a value of $t/r = 0.0185$, the maximum allowable localized buckling stress due to bending moment is $\sigma_{\text{buckling, bending}} = 191,000$ psi, which is comparable to the yield stress $\sigma_{\text{yield}} = 170,000$ psi.

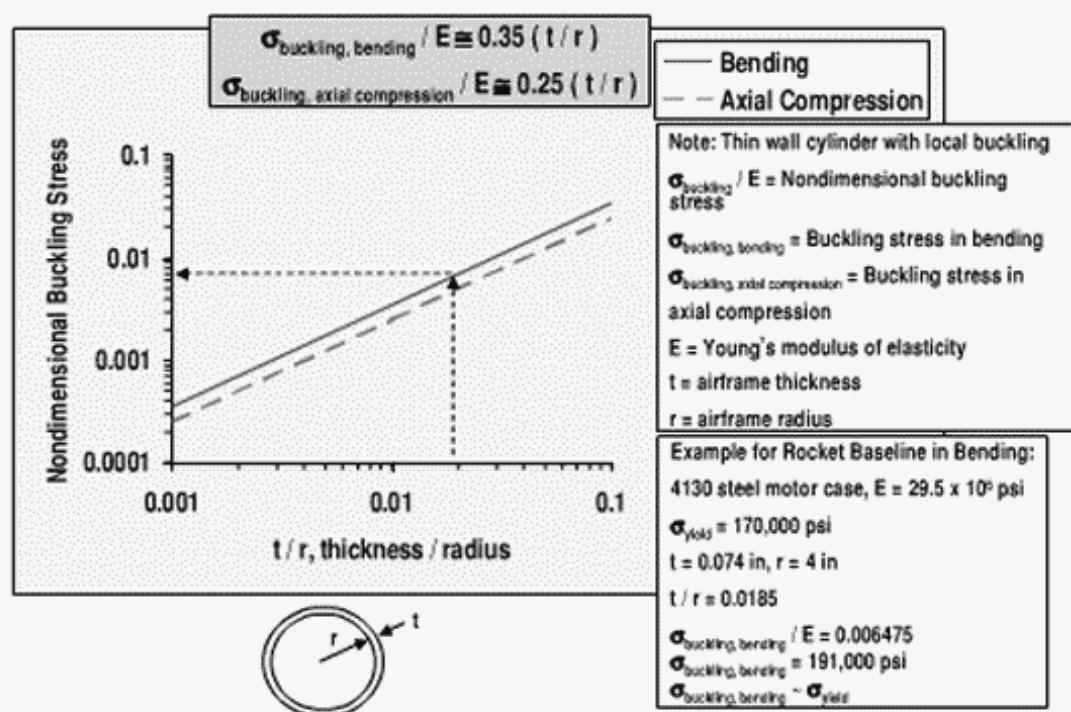


Fig. 4.20 Localized buckling for thin-wall missile structure.

Flight carriage impact on missile design is illustrated in Fig. 4.21. A comparison is shown of a representative distribution of missile free-flight maneuver loads vs launch platform carriage loads. The left section of the figure shows a typical free-flight maneuvering air-load distribution and the weight-load distribution on each bulkhead. The right section of the figure shows a typical maneuvering aircraft carriage air-load distribution, carriage weight-load distribution, and carriage suspension loads. The missile free-flight loads are usually higher than the carriage loads because missile maximum maneuverability is usually greater than that of aircraft (e.g., 30 g vs 9 g). The missile skin thickness is usually not sized by aircraft maneuver loads. As shown in the right section of the figure, carriage loads are taken out through a suspension system. It is usually possible to get a fairly accurate prediction of the missile free-flight loads. Also, wind-tunnel tests are usually conducted to determine free-flight air loads. Unfortunately, this is usually not the case for carriage flight loads, as it is difficult to accurately predict the two-body problem of a store in the flow field of the launch aircraft. In addition, it is difficult to get accurate wind-tunnel data because of the small size of the missile model for aircraft carriage wind-tunnel tests. As a result, the current approach to estimating carriage loads is usually based on the conservative process of Military Standard (MIL STD) MIL-A-8591. As missile loads estimation becomes more accurate in the future, there is a potential for structure weight savings based on improving the estimation accuracy for carriage loads.

Shown in Fig. 4.22 is a nomogram that calculates the bending moment M_B as a function of force N , length l , and the type of loading c . The scales of the nomogram cover the typical maximum/minimum ranges of the values of the parameters for tactical missiles. An advantage of using a nomogram is insight into the design drivers.

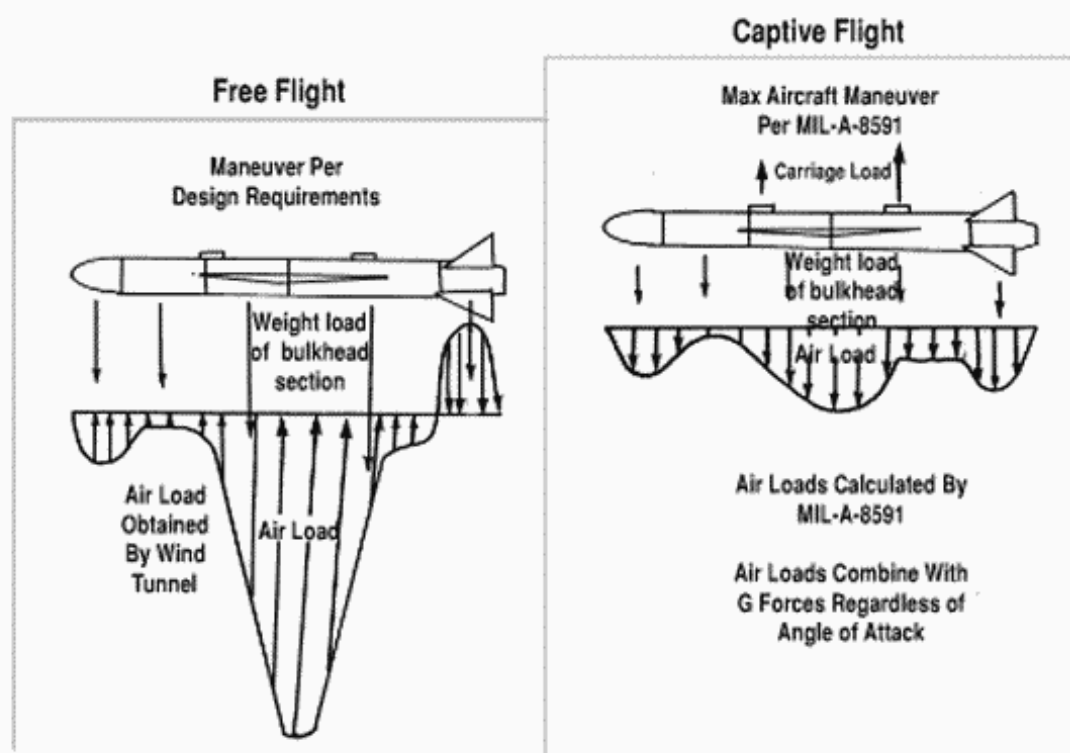


Fig. 4.21 Process for captive and free-flight loads calculation.

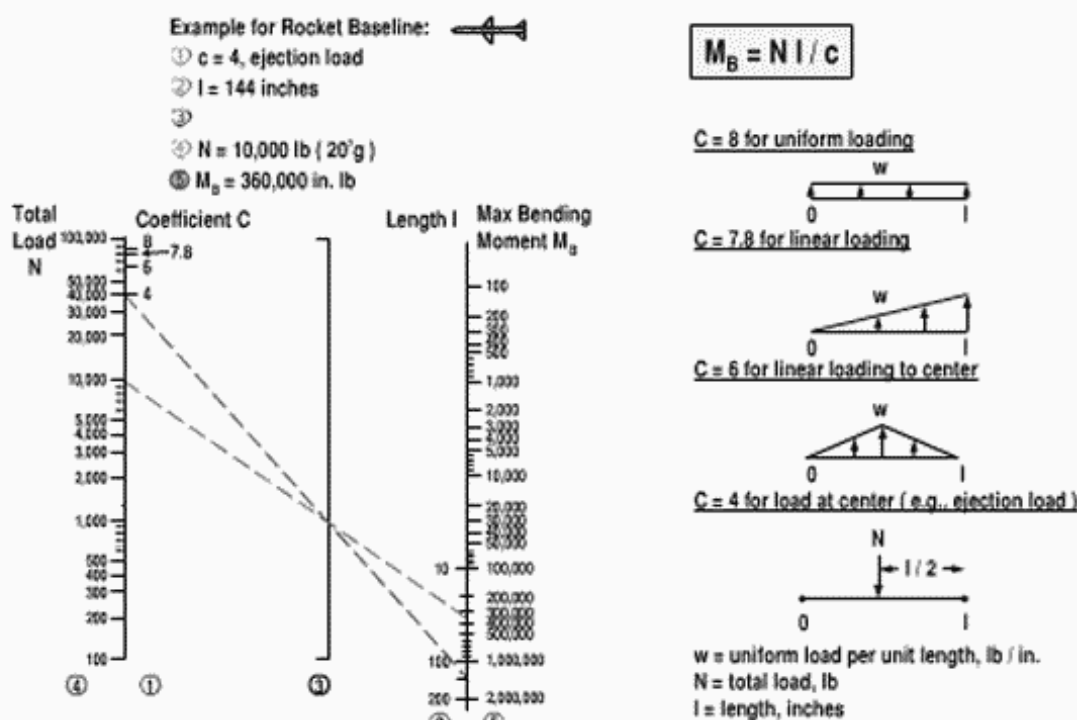


Fig. 4.22 Maximum bending moment depends upon load distribution.

Referring again to the figure, note the shape of the typical load distributions. The load distributions are uniform loading ($c = 8$), linear loading increasing from the nose tip to the base ($c = 7.8$), linear loading increasing from the nose tip to the center of the missile and returning to zero at the base ($c = 6$), and point loading (e.g., ejection load) at the center ($c = 4$).

As an example for the rocket baseline missile subjected to an ejection load of 20 g (10,000 lb), the bending moment is calculated as $M_B = Nl/c = 10,000(144)/4 = 360,000$ in. · lb. The bending moment is estimated in the nomogram by drawing a straight line from line 1 ($c = 4$) to line 2 ($l = 144$) to determine a turning point on line 3. A second straight line is drawn from line 4 ($N = 10,000$) through the turning point on line 3 to intersect the value of the bending moment on line 5 ($M_B = 360,000$ in. · lb).

The simple nomogram shown in the figure is a throwback to the way engineering estimates were made more than 30 years ago, before the advent of high-speed computers. Although nomograms are still in use in areas such as medicine, they have fallen out of use in aerospace engineering. Until about the year 1970 nomograms, including complex special slide rules with three or more sliders, were used in aerospace design and as an aerospace engineering teaching tool in universities.

An example is shown in Fig. 4.23 of the required thickness of the rocket baseline missile airframe to maintain bending moment stress below the yield stress. The required thickness is calculated as a function of bending moment, airframe radius, maximum allowable stress, and factor of safety. The equation is $t = M_B(\text{FOS})/[\pi r^2 \sigma_{\text{Max}}]$. Assumptions for the equation are a thin cylinder of circular cross section, solid skin, longitudinal bending moment, negligible axial (thrust and drag) stress, and negligible thermal stress. As noted in the previous discussion, the rocket baseline subjected to an ejection load of 20 g will have a bending moment $M_B = 360,000$ in. · lb. Based on a factor of safety for an ultimate load

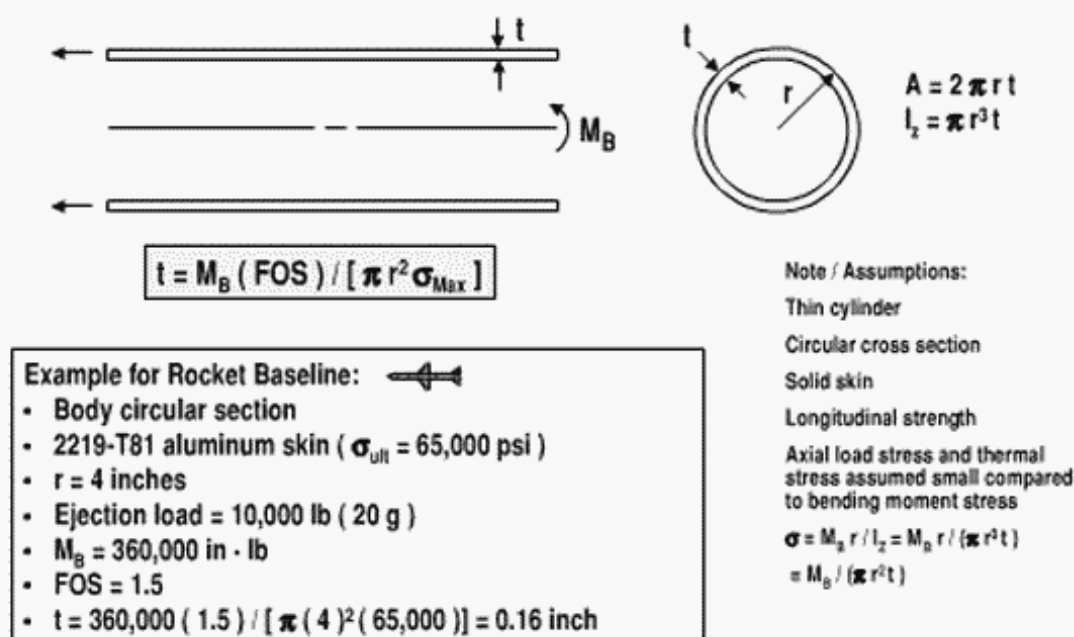
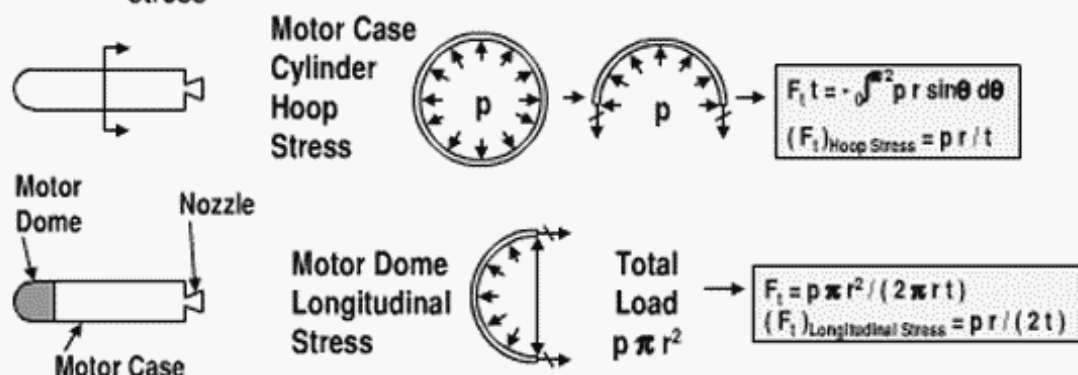


Fig. 4.23 Bending moment load may drive body structure design.

- ♦ For an axisymmetric motor case with a hemispherical dome, motor case cylinder hoop stress is twice the motor dome longitudinal stress



- ♦ With metals – the extra longitudinal material reacts body bending
- ♦ In filament wound motor designs, extra fibers must usually be added to accommodate body bending

Fig. 4.24 Motor case stresses may be determined from the pressure vessel.

of $FOS = 1.5$, an aluminum airframe structure ultimate stress $\sigma_{ult} = 65,000$ psi, and an airframe radius $r = 4$ in., the required airframe thickness is computed to be $t = 0.16$ in.

The rocket motor case is subjected to a variety of loads. Determining the type of load and its value that sizes the motor case requires an evaluation of a variety of loads. The loads on the motor case include internal pressure from the gas produced during propellant combustion; thrust produced by the motor during operation; localized buckling, aerodynamic bending, and the ground handling loads developed during a variety of flight conditions (e.g., handling, storage, launch platform carriage, ejection, and flight); and thermal expansion stress.

The dominant load on the motor case is often due to internal pressure. Figure 4.24 illustrates the motor case stress from the internal pressure. The motor case hoop stress is a function of the internal pressure p , case thickness t , and case radius r . Hoop stress is the radial stress in the cylindrical section of the motor case. It is given by the equation $(F_t)_{\text{Hoop Stress}} = p r / t$. The motor case longitudinal stress is the stress in the forward dome. For a hemispherical dome it is one-half of the hoop stress and is given by $(F_t)_{\text{Longitudinal Stress}} = p r / (2 t)$.

Because the motor case cylindrical section and the forward dome section are often the same thickness, a dome thickness set by internal pressure often has additional material, providing additional design margin to react to other loads such as body bending loads. In the case of a composite motor case, the case thickness may not be set by internal pressure but by body bending loads, requiring a greater thickness to accommodate body bending.

The following example calculates the required thickness of the steel motor case of the rocket baseline as a function of pressure, factor of safety, radius, and the maximum allowable stress. The calculation shows the weight savings of a composite motor case over a steel motor case.

Step 1. Calculate maximum effective operating pressure (MEOP):

- 1) $MEOP = (OP)_{RT} e^{\pi_k \Delta T}$ (margin for ignition spike and other design uncertainty).
- 2) Rocket baseline: motor diameter = 8 in., motor length = 55 in.
- 3) $(OP)_{RT}$ = operating pressure at room temperature = 1769 psi average boost pressure at room temperature (RT).
- 4) $\pi_k = (\Delta p / \Delta T) / p_c = 0.14\% / ^\circ F$, where π_k = sensitivity of motor pressure to temperature.
- 5) Assume hot day = $140^\circ F$, calculate $e^{\pi_k \Delta T} = e^{0.0014(140-70)} = 1.10$.
- 6) For a 3σ design margin, $MEOP = 1769 \times (1.10)^3 = 2355$ psi.

Step 2. Assume an ultimate factor of safety $FOS = 1.5$.

Step 3a. First try the rocket baseline steel material for the motor case. The maximum allowable tensile stress is $F_t = 200,000$ psi ultimate. Calculate:

- 1) $t_{Hoop} = (MEOP) \times r \times (FOS) / F_t = 2355 \times 4.0 \times 1.5 / 200,000 = 0.071$ in.
- 2) $t_{Dome} = 0.035$ in.
- 3) Weight = $W_{Cylinder} + W_{Dome} = \rho \pi d t_{Hoop} \ell + \rho (\pi/2) d^2 t_{Dome} = 29.9$ lb for a steel motor case (without attachments and insulation).

Step 3b. Next, try an alternative composite glass/epoxy material for the motor case. The maximum allowable stress in the glass is $F_t = 450,000$ psi ultimate. Assume 60% glass/40% epoxy. Calculate:

- 1) $t_{Hoop} = 2355 \times 4.0 \times 1.5 / [(450,000)(0.60)] = 0.052$ in.
- 2) $t_{Dome} = 0.026$ in. May need to add additional thickness to react body bending load.
- 3) Weight = 10.1 lb for composite case (one-third of the weight of the rocket baseline steel motor case).

4.13 Seeker Dome Materials

Shown in Fig. 4.25 is a comparison of alternative dome materials for missile seekers. The dome materials are grouped based on their best applicability to multi-mode (RF/IR), RF-only, and mid-wave IR-only seekers. Measures of merit are dielectric constant, combined mid-wave/long wave infrared bandpass, transverse strength, thermal expansion, erosion resistance, and maximum short-duration temperature. Dome materials that are especially suited for combined radar and infrared seekers are zinc sulfide and zinc selenide. Zinc sulfide has advantages in dielectric constant, transverse strength, and rain erosion. Zinc sulfide is generally the multimode dome material of choice for Mach numbers up to 3. For Mach number greater than 3, new materials are required for multimode seekers. Candidate materials include spinel/sapphire, quartz/fused silicon, and silicon nitride. These materials are more expensive than zinc sulfide and zinc selenide. A new candidate dome material that is under development for missile defense applications is diamond. Obviously cost is very high for a diamond dome. In addition to high material cost, diamond dome assembly cost is high. Diamond domes must be assembled as a built-up mosaic because the present manufacturing processes produce relatively small size diamonds.

For RF-only seekers, two popular radome materials are pyroceram and polyimide. Pyroceram is commonly used in supersonic missiles, such as Sparrow and the Patriot PAC-2. Polyimide radomes are used on relatively low speed, low cost

Seeker Dome Material	Density (gm / cm ³)	Dielectric Constant	MWIR / LWIR Bandpass	Transverse Strength (psi)	Thermal Expansion (10 ⁻⁶ / °F)	Erosion, Knoop (kg / mm ²)	Max Short-Duration Temp (°F)
RF-IR Seeker							
Zinc Sulfide (ZnS)	4.05	○ 8.4	●	○ 18	○ 4	○ 350	○ 700
Zinc Selenide (ZnSe)	5.16	◐ 9.0	●	◐ 8	○ 4	○ 150	○ 600
Spinel (MgAl ₂ O ₄)	3.68	○ 8.5	◐	○ 28	○ 3	● 1650	● 1800
Quartz / Fused	2.20	● 3.7	○	◐ 8	● 0.3	◐ 600	● 2000
Silicon (SiO ₂)							
Silicon Nitride (Si ₃ N ₄)	3.18	◐ 6.1	◐	● 90	◐ 2	● 2200	● 2700
Diamond (C)	3.52	◐ 5.6	◐	● 400	● 1	● 8800	● 3500
RF Seeker							
Pyroceram	2.55	◐ 5.8	◐	○ 25	○ 3	◐ 700	● 2200
Polyimide	1.54	● 3.2	◐	○ 17	◐ 40	◐ 70	◐ 200
MWIR Seeker							
Magn. Fluoride (MgF ₂)	3.18	◐ 5.5	●	◐ 7	◐ 6	◐ 420	◐ 1000
Alon (Al ₂ O ₃ :N ₅)	3.67	◐ 9.3	◐	◐ 44	○ 3	● 1900	● 1800

● Superior ◐ Above Average ○ Average ◐ Below Average

Fig. 4.25 Dome material selection is driven by the type of seeker.

missiles such as the millimeter wave (mmW) Brimstone. Polyimide radomes have excellent dielectric characteristics. For MWIR-only seekers, additional dome materials include magnesium fluoride and Alon. Although both are suitable for supersonic missiles, Alon is less susceptible to rain and dust erosion and is capable of operating at higher Mach number. Multispectral dome materials may also be used for MWIR-only and RF-only seekers. Zinc sulfide is suitable for MWIR seekers at supersonic Mach number. Spinel or sapphire domes may be used with MWIR seekers at high supersonic/low hypersonic Mach numbers. Silicon nitride is suitable for RF and mmW seekers at low hypersonic Mach number.

4.14 Thermal Stress

The thermal stress trouble spots shown in Fig. 4.26 are the dome, joints, and leading edges. Radomes and IR domes often have thermal stress problems at the attachment of the dome to the metal airframe because of a difference in thermal expansion. Domes have their thickness set by electromagnetic or electro-optical (EO) considerations in addition to structural considerations. Because many dome materials have low strength, domes are susceptible to failure.

Body joints are also susceptible to local stress concentration. For a supersonic or hypersonic missile, the difference in temperature of the hot outer airframe (due to aerodynamic heating) and the relatively cold inner bulkhead (due to the thermal heat sink of the interior bulkhead material) may lead to high local thermal stress at the body joint. This may cause premature buckling.

Finally, leading edges contribute to high aerodynamic heating for hypersonic missiles. The leading edges are subjected to stagnation heating and could erode. Downstream of the hot leading edge, the additional thickness of the wing, canard, or tail surface is relatively cold. The difference in temperature may cause the

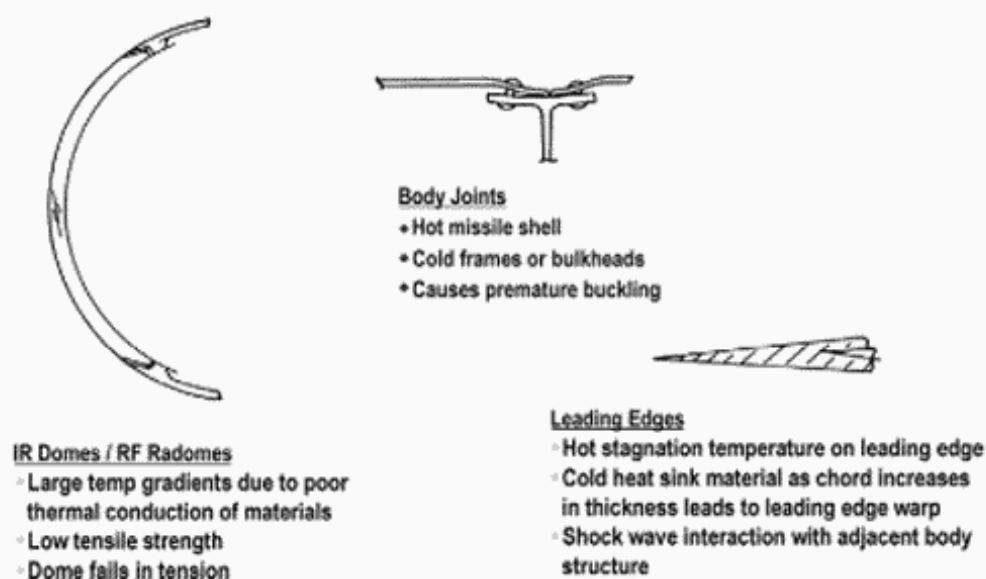


Fig. 4.26 Thermal stress trouble spots.

leading edge to warp. Another concern is the impingement of the shock wave from the leading edge on an adjacent surface. Shock-wave/boundary-layer interaction causes localized high heating.

4.15 Localized Aerodynamic Heating

Shown in Fig. 4.27 are examples of aerodynamic hot spots for a supersonic or hypersonic tactical missile. Note the high aerodynamic heating at the nose tip, leading edge of surfaces, corner flow at the cylinder-flare junction, and the reattachment of the separated boundary layer downstream of the cylinder-flare junction. The aerodynamic heating is higher on the windward side of the missile. Note

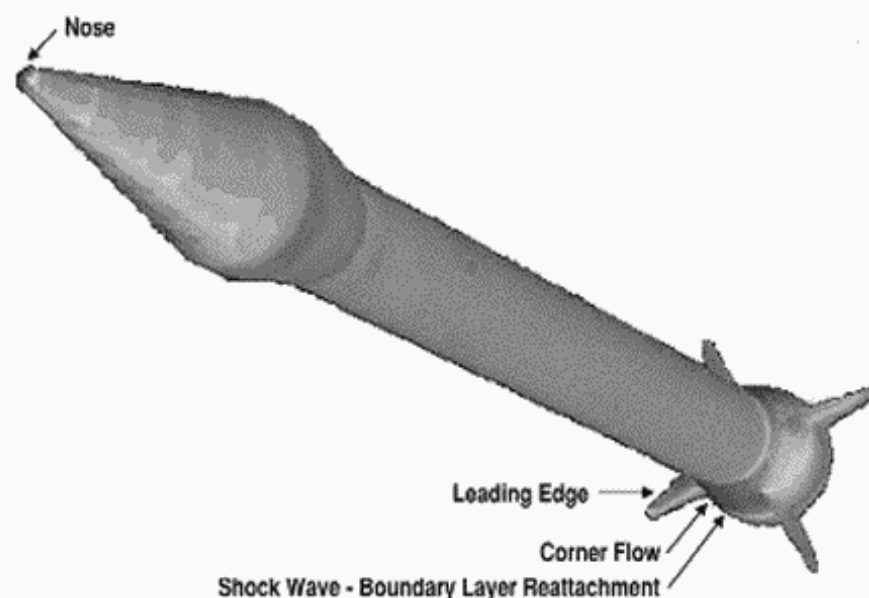


Fig. 4.27 Examples of aerodynamic hot spots.

that the expansion region aft of the bulge is a relatively cool region of low aerodynamic heating. The aerodynamic hot spots may require either additional local insulation or the use of a high temperature airframe material to withstand the high local heating.

4.16 Summary

This chapter addressed design methods, manufacturing processes, design considerations, and the technologies for lightweight, low-cost tactical missiles. Design methods included weight prediction, center-of-gravity and moment-of-inertia prediction, factors of safety, aerodynamic heating prediction, and structure design methodology. Manufacturing processes include low parts count/low-cost processes such as precision castings, vacuum assisted resin transfer molding, pultrusion, extrusion, and filament winding. Design considerations included airframe material alternatives, insulation material alternatives, seeker dome material alternatives, thermal stress, and aerodynamic heating. Finally, new technologies were identified for lighter weight missiles. These included MEMS diagnostic technology, composites, titanium alloys, and higher density insulation.